Gated Context Memory

Paper A Research Brief

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Long-Context Memory

July 22, 2026

Paper A research brief: July 2026, Week 3
Status through Wed 2026-07-22 06:33 PT

Research question and method

Question

When can an agent answer from a short learned state, and when should it return to raw evidence?

1. Recurrently encode context chunks into soft memory while carrying earlier memory forward.
2. Read memory through a small LoRA adapter on the same frozen language model.
3. Keep an adapter-free bounded raw path on the same backbone.
4. Use held-out confidence to study routing between the two paths.

Positioning

Compression ratio alone is not the contribution. Paper A studies the complete **compact-memory versus raw-evidence interface**, including paired failure, coverage, and cost.

Core result: useful compact state, clear boundary

base	method	QuALITY	BFCL	HotpotQA
Qwen3-8B	Raw ref.	7.2	92.4	53.7
	SFT ref.	81.7	95.4	68.8
	LLMLingua-2	14.3	70.3	22.1
	Compressor (w/o gate)	54.4	72.3	28.9
	Compressor (w/ gate)	**54.6**	**88.5**	**50.9**
Qwen3.5-9B	Raw ref.	7.1	84.5	53.9
	SFT ref.	85.0	94.9	71.7
	LLMLingua-2	20.3	60.8	28.9
	Compressor (w/o gate)	**51.5**	72.0	30.5
	Compressor (w/ gate)	51.4	**80.5**	**51.7**

- The compressor without the gate is the strongest completed compact path in the core table.
- SFT is a full-cost adaptation reference, not a compression competitor.
- Formal testing certifies no nonzero coverage; routing is reported as held-out empirical analysis.

Real long-context transfer

base	method	LongBench-v2	∞Bench-choice	BABILong
Qwen3-8B	Raw ref.	31.2	52.8	18.1
	Compressor (w/o gate)	20.1	21.4	0.9
	Compressor (w/ gate)	**31.4**	**52.8**	**18.1**
Qwen3.5-9B	Raw ref.	33.0	52.4	10.5
	Compressor (w/o gate)	22.5	31.9	2.2
	Compressor (w/ gate)	**33.4**	**52.4**	**11.9**

Task-structured operating region

The compressor without the gate can beat truncated raw on multi-hop LongBench transfer, while raw wins on single-document / extractive targets. The gate selects between the two paths: **Compressor (w/ gate) $\geq$ Raw on every harvested two-base target**.

Single seed; error bars, RULER, and K32 repairs remain.

Execution status and next steps

stage	done	run	fail	pending
core main	88	0	0	0
replicate + SFT audit	9	0	0	0
transfer adapters	46	0	2	0
real long-context	50	3	2	63
generality	11	1	0	36
budget / RULER	0	0	0	23
ablation	0	0	0	36
official baselines	0	0	0	52

Current workers

- 3 long-context runs;
- 1 generality run;
- 1 unavailable GPU.

Order to completion

1. continue long-context;
2. generality;
3. RULER / budget;
4. ablation + cost;
5. official baselines.